Chapter 13 - Eigenvalues

Nonhomogeneous Equations:

$Ax = b$ - if eq^{n}s are independent $\Rightarrow$ a unique solution

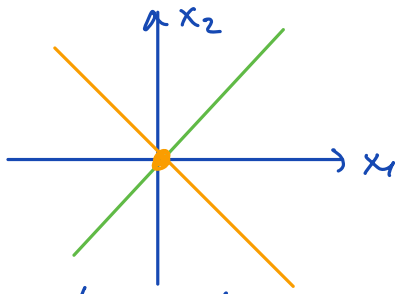
Homogeneous linear system

$Ax = 0$

if eq^{n}s are independent:

$a_{11}x_1 + a_{12}x_2 = 0$

$a_{21}x_1 + a_{22}x_2 = 0$



Two lines intersecting at $x_1 = x_2 = 0$.
Trivial solution : all x's $= 0$

Eigenvalue Problems

$Ax = \lambda x$ $\Rightarrow$ Solve for x's: eigenvectors of A

" " λ's: eigenvalues of A

$Ax = \lambda I x$ ↳ identity matrix

$(A - \lambda I)x = 0$, Goal: Find λ, x that makes left hand side zero.

For the nontrivial solution to exist, the rows or colums of the matrix $(A - \lambda I)$ has to be linearly dependent.

$\Rightarrow \quad |A - \lambda I| = 0$

expand the determinant $\Rightarrow$ get a polynomial in λ (characteristic polynomial)

e.g. Two eq^n case:

$$A - \lambda I = \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$$

$$|A - \lambda I| = 0 \quad \Rightarrow \quad (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0$$

$$\lambda^2 - (a_{11} + a_{22})\lambda - a_{12}a_{21} + a_{11}a_{22} = 0$$

$$\lambda_{1,2} = \frac{(a_{11} + a_{22}) \pm \sqrt{(a_{11} + a_{22})^2 - 4(\dots)}}{2}$$

EX $\left.\begin{aligned} 10x_1 - 5x_2 &= 0 \\ -5x_1 + 10x_2 &= 0 \end{aligned}\right\} \Rightarrow (A-\lambda I)x = 0:$

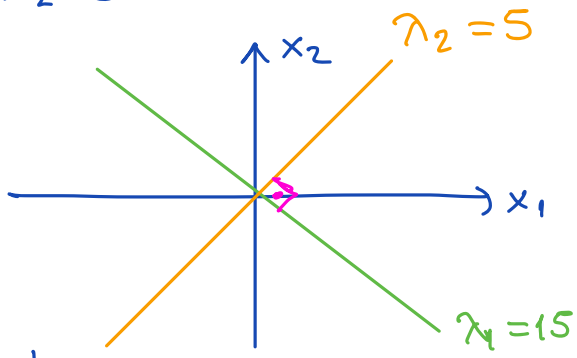
$(10-\lambda)x_1 - 5x_2 = 0 \quad (1)$

$-5x_1 + (10-\lambda)x_2 = 0 \quad (2)$

$$|A-\lambda I| = \begin{vmatrix} 10-\lambda & -5 \\ -5 & 10-\lambda \end{vmatrix} \Rightarrow \lambda^2 - 20\lambda + 75 = 0$$

$\lambda_1 = 15$

$\lambda_2 = 5$



Put λ_1 into (1) and (2):

$\left.\begin{aligned} -5x_1 - 5x_2 &= 0 \\ -5x_1 - 5x_2 &= 0 \end{aligned}\right\}$ identical

$\downarrow$

$x_2 = -x_1$

One solution: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ $\rightsquigarrow$ eigenvector $= \underline{e}_1$

Put $\lambda_2 = 5$ into (1) & (2):

$\left.\begin{aligned} 5x_1 - 5x_2 &= 0 \\ -5x_1 + 5x_2 &= 0 \end{aligned}\right\}$ identical $\Rightarrow \infty$ solutions

$x_1 = x_2$ choose $x_1 = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\text{eigenvector.}} = \underline{e}_2$

- Eigenvectors are <u>orthogonal</u>
  (a property of symmetric matrices) (A is symmetric:

$$A = \begin{bmatrix} 10 & -5 \\ -5 & 10 \end{bmatrix}$$

- $\underbrace{Ax = \lambda x}$

information in A is translated to λ and eigenvectors.